Cloudworx Industrial Digital Twin Platform

Architecture Design Document — Surface 1 & Surface 2 | Reference: CWX-ARCH-2026

Authors: Senior Frontend Architect & 3D Web Graphics Specialist

Product Surfaces: Surface 1 (Procurement Workspace) & Surface 2 (Embedded 3D Configuration Workspace)

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1. Executive Summary

This architecture design document establishes the technical blueprint for the Cloudworx Industrial Digital Twin platform across two core product surfaces:

1. **Surface 1: Enterprise Procurement Workspace:** An enterprise 2D web application serving 3,000 global buyers and suppliers across Vietnam, India, and China for supplier onboarding, document compliance, and multi-line item RFQ comparisons over high-density data tables (up to 50,000 suppliers and 200,000 document records).
2. **Surface 2: Embedded 3D Configuration Workspace:** An industrial 3D micro-application embedded within an OEM partner's production product-selection host portal. It enables sales engineers and plant architects to layout industrial electrical rooms (20m × 12m envelope), position drives, cabinets, and control panels with 0.5m grid quantization, observe real-time engineering clearance rules (< 0.6m separation), inspect live bills of materials (BOM), and collaborate with an AI assistant that previews and proposes automated layout modifications.

Scope Distinction: Implemented Prototype vs. Proposed Production Architecture

Capability	Part B Implemented Prototype	Proposed Production Architecture
Framework & Language	React 18, TypeScript (Strict), Vite 6	React 18, TypeScript Monorepo (pnpm workspaces / Turborepo)
3D Engine	Three.js via React Three Fiber (R3F) & Drei	R3F + Three.js + Custom WebGPU / WebGL2 Fallback Pipeline
State & Edit History	Zustand store with snapshot-based undo/redo (max 50)	Zustand for local scene + TanStack Query for server sync + WebSocket CRDT
Grid & Asset Scale	20m × 12m grid, 3 primitive types, 500 items benchmark	300 glTF CAD models, 3 LODs, Draco/KTX2, up to 2,000 items
Rule Validation	In-memory AABB edge-to-edge 0.6m clearance check	Spatial hash grid client-side + Server-side asynchronous validation
Host Embedding	In-app PostMessage simulator modal (origin + payload checks)	Sandboxed iframe + Web Component MFE option + CSP Level 3
AI Assistant	Deterministic mock with streaming generator & ghost preview	Streaming LLM service via Server-Sent Events (SSE) with JSON Schema
Localisation	English & Vietnamese (react-i18next)	English, Vietnamese, Simplified Chinese (i18next + ICU message format)

2. Requirements and Constraints Analysis

Key Constraints and Architecture Drivers

1. **Host-Owned Authentication (OpenID Connect / OAuth2):**

• *Impact:* The embedded 3D application cannot maintain an independent login portal. Authentication context (bearer token, tenant ID, user role) is passed across the boundary via a secure `postMessage` handshake. The micro-app must

gracefully operate in degraded mode if host credentials expire.

2. **Hardware Constraints (Corporate Laptops with Integrated Graphics & 8 GB RAM):**

• *Impact:* Corporate users run locked-down Chrome or Edge, often inside VDI or over 10 Mbps VPNs. The 3D scene must achieve ≥ 50 FPS at 1080p, with < 300 draw calls and < 1 GB browser memory. This mandates aggressive geometry and material instancing, texture compression, and strict memory deallocation.

3. **Data Scale (Grids up to 50k Suppliers & 200k Documents):**

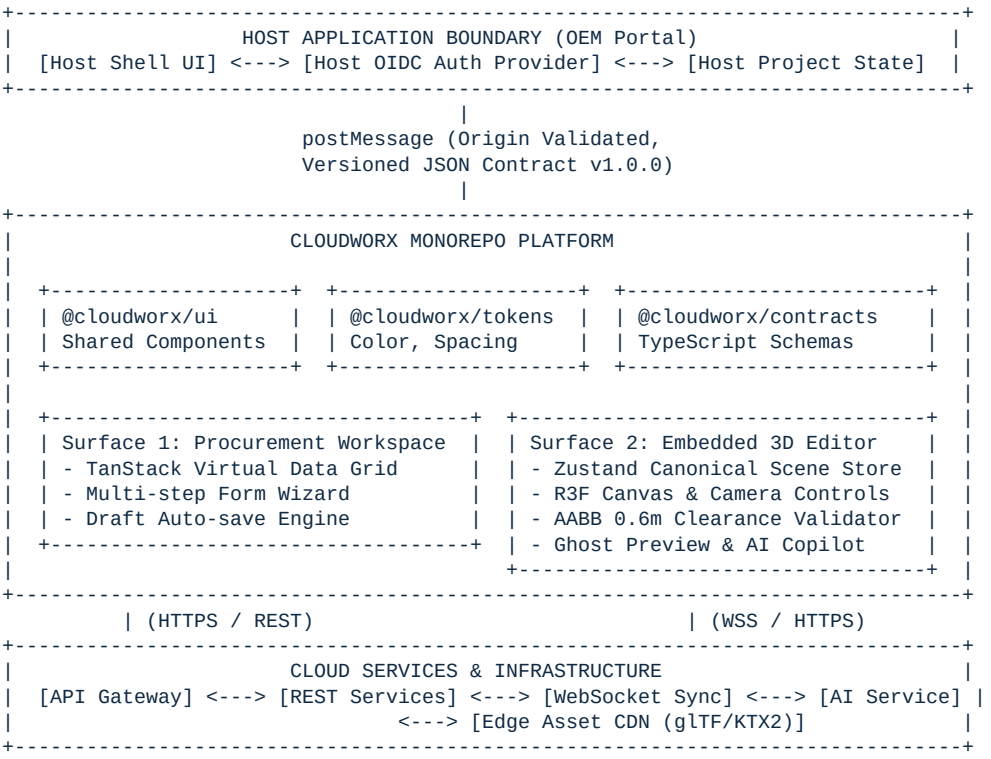
• *Impact:* Rendering massive datasets in the Procurement Workspace requires DOM virtualization (rendering only visible rows + overscan) and cursor-based server pagination. Traditional un-virtualized DOM tables will freeze the browser tab.

4. **Accessibility (WCAG 2.2 AA) & Internationalisation:**

• *Impact:* One unified codebase must support English, Vietnamese, and Simplified Chinese, including locale-specific currency (USD, VND, CNY, INR) and number formatting. The 3D canvas must have an accessible DOM sibling (Scene Hierarchy List) allowing full keyboard selection, inspection, translation, and deletion.

3. System Architecture Diagram

The system topology separates host boundaries, shared monorepo packages, UI layers, local state engines, and cloud microservices.



(Source definition maintained in `docs/architecture-diagram.mmd`*)*

4. Application Structure and Module Boundaries

We propose an enterprise monorepo managed with **pnpm workspaces** and **Turborepo**:

```
cloudworx-monorepo/
|-- apps/
|   |-- procurement/              # Surface 1: Enterprise Procurement SPA
|   |   |-- src/features/onboarding/
|   |   |-- src/features/compliance/
|   |   +-- src/features/rfq-compare/
```

```
|   +-- layout-editor/            # Surface 2: Embedded 3D Configuration Workspace
|       |-- src/features/editor/
|       |-- src/features/palette/
|       |-- src/features/properties/
|       +-- src/features/assistant/
+-- packages/
    |-- contracts/                # Shared TypeScript interfaces & Zod schemas
    |-- design-tokens/            # Semantic tokens (CSS variables & Tailwind)
    |-- ui/                       # WCAG 2.2 AA component library
    |-- i18n/                     # Core i18next engines & shared translation keys
    +-- tsconfig/                 # Shared base TypeScript configurations
```

Decoupled Release Trains

- The Procurement Workspace and 3D Layout Editor are built as independent artifacts and deployed to separate CDN endpoints.
- Breaking changes in `@cloudworx/contracts` utilize semantic versioning. The 3D micro-app can be updated and released without triggering a redeploy of the Procurement application.

5. Technology Selection and Justification

Technology	Selected For	Justification	Alternatives Rejected
React 18 + TS	Both Surfaces	Fixed assessment requirement; strict type safety prevents runtime regressions in complex geometric calculations.	Plain JS (untenable for large CAD models).
Vite 6	Build & Dev Tool	Sub-second HMR, native ES module compilation, superior Rollup chunking.	Webpack (slower cold starts, complex config).
React Three Fiber (R3F)	3D Viewport	Declarative scene graph that reconciles directly into Three.js without wrapper overhead.	Direct imperative Three.js (spaghetti code), Babylon.js (heavy bundle).
Zustand	Editor State	Lightweight (< 1.5 KB), unopinionated, zero boilerplate, outside-React selector subscriptions.	Redux Toolkit (boilerplate overhead), Context API (re-render cascades).
TanStack Query	Server State	Automatic background caching, query invalidation, retry backoff, and optimistic rollback.	SWR (fewer mutation primitives).
TanStack Virtual	Data Grids	Virtualizes DOM rows to keep DOM node count constant (~30 nodes for 200k rows).	Un-virtualized HTML table (crashes browser).
i18next	Internationalisation	Built-in namespace support, interpolation, pluralization, locale formatting.	FormatJS / react-intl (heavier runtime).
Vitest	Testing Suite	Native Vite integration, fast parallel test execution, compatible with Jest APIs.	Jest (slow Babel transformation overhead).

6. State and Data Architecture

State Separation Strategy

```
                        STATE TAXONOMY
  +-------------------------------------------------------------+
  | 1. Server State (TanStack Query)                            |
  |    - Project metadata, saved room configurations, BOM catalogs|
  +-------------------------------------------------------------+
  | 2. Canonical Scene State (Zustand)                         |
  |    - items: Record<string, PlacedItem>                      |
  |    - Serialisable, immutable, single source of truth        |
  +-------------------------------------------------------------+
  | 3. Undo/Redo Engine (Zustand Snapshot History)              |
  |    - history: SceneSnapshot[], historyIndex: number         |
  |    - Isolated from transient selection & viewport state     |
  +-------------------------------------------------------------+
  | 4. Ephemeral / Transient UI State                           |
  |    - selectedItemId, hoveredItemId, activeMode, camera matrix|
  +-------------------------------------------------------------+
  | 5. AI Assistant Proposal State                              |
  |    - aiProposal: { itemId, newX, newZ, status }            |
  |    - Visualized via GhostPreview before user confirmation   |
  +-------------------------------------------------------------+
```

Offline & Degraded Mode on Flaky VPNs

- **Optimistic Mutations:** When saving a layout, the UI updates immediately and queues an HTTP `PUT` request with exponential retry backoff.
- **WebSocket Reconnection Protocol:** Upon VPN reconnect, the editor sends a `RESYNC_SCENE` message with its last known vector clock timestamp to resolve concurrent multi-user conflicts (Last-Write-Wins with property-level merging).

7. High-Density Data Grids and Long Multi-Step Forms

Large Grids (50,000 Suppliers / 200,000 Documents)

- **DOM Virtualization:** Only 25–35 visible rows are mounted in the DOM at any instant. A scroll container of simulated height (N × rowHeight) translates items via CSS `transform: translateY()`.
- **Server-Side Pagination:** Keyset/cursor pagination (`cursor=supplier_9941&limit=50`) avoids expensive database offset scans.
- **Saved Views & Filtering:** Filter predicates are stored in URL query parameters, enabling direct bookmarking and shareable links.

Multi-Step Form Persistence & Recovery

- Long forms (e.g. 5-step Supplier Compliance Verification) persist partial drafts to `IndexedDB` on every input blur.
- If the browser tab crashes or the VPN disconnects, the user is presented with a "Resume Draft" prompt upon reload.
- Server-side validation returns standard RFC 7807 problem details mapped to individual form fields.

8. Internationalisation and Accessibility (WCAG 2.2 AA)

Internationalisation (i18n)

- **Languages:** English (`en-US`), Vietnamese (`vi-VN`), and Simplified Chinese (`zh-CN`).
- **Locale Formatting:** Uses `Intl.NumberFormat` and `Intl.DateTimeFormat` for currency display (USD, VND, CNY, INR) and number separators.
- **CJK Line Breaking & Expansion:** CSS `word-break: break-word` and `overflow-wrap: anywhere` prevent clipping when Vietnamese text expands by 25–35% compared to English.

Accessibility in 3D Environments

- 3D WebGL canvases are non-semantic to screen readers.
- **Accessible Sibling Pattern:** The 3D editor provides a synchronized DOM Scene Hierarchy (`src/features/scene-list/SceneList.tsx`) with `role="listbox"` and `aria-selected` attributes.
- **Keyboard Navigation:** Users can navigate items via Up/Down keys, nudge items by 1 grid cell (0.5m) using Arrow keys, rotate items via `R`, and delete via `Delete`, with live `aria-live="polite"` clearance violation announcements.

9. Host Embedding and Security

Iframe vs. Micro-Frontend Evaluation

Aspect	Iframe Architecture (Recommended for Pilot)	Micro-Frontend / Module Federation
CSS & JS Isolation	Complete sandboxing; zero style bleed or global namespace collisions.	Shared runtime; styles can leak without strict Shadow DOM.
Security Boundary	High; strict origin checking, CSP `frame-ancestors`.	Low; host and micro-app share same DOM and JS memory.
Context Communication	Explicit via `window.postMessage`.	Shared memory / JS Event Target.
Recommendation	**Primary architecture for Surface 2**	Evaluated for enterprise hosts that prohibit iframes

Versioned PostMessage Handshake Protocol (v1.0.0)

1. **Host Initialization (`CLOUDWORX_HOST_INIT`):**

```
{
  "type": "CLOUDWORX_HOST_INIT",
  "version": "1.0.0",
  "payload": {
    "projectId": "PRJ-VN-2026-0891",
    "user": { "id": "usr-1", "role": "SalesEngineer" },
    "roomDimensions": { "width": 20, "depth": 12 }
  }
}
```

2. **Security Checks:**

- Verify `event.origin === hostOrigin` (never allow wildcard *).
- Verify `event.source === window.parent`.
- Validate payload against Zod schema.

3. **Token Handling:** Bearer tokens are stored exclusively in in-memory closures—never in `localStorage` or URL query params.

10. 3D Editing Architecture

- **World Coordinate Space:** Cartesian origin (0, 0, 0) at room corner. +X extends across floor width (0 to 20m), +Z extends across room depth (0 to 12m), +Y points vertically upwards.
- **Snapping Quantization:**

$snap(v) = round≤ft(\frac{v}{0.5}\right) × 0.5$

- **Boundary Clamping:** Item coordinates are clamped to [W_{half}, 20 - W_{half}] and [D_{half}, 12 - D_{half}].
- **Reversible Edit Commands:** Every add, move, rotate, delete, or accepted AI proposal creates a snapshot on the history stack, enabling full `Ctrl+Z` / `Ctrl+Shift+Z` navigation.

11. 3D Asset Optimisation Pipeline

Converting a 40 MB raw CAD export (STEP/IGES) into a web-ready asset:

```
[Raw CAD STEP/IGES (40 MB)]
          |
     Meshing & NURBS tessellation
          |
[Unoptimized glTF (25 MB, 350k polys)]
          |
     gltf-transform (meshopt, deduplication, weld)
          |
     Mesh Simplification (3 LOD levels: 100%, 25%, 5%)
          |
     Texture Compression (basisu -> KTX2 / UASTC)
          |
[Production Web Asset (1.2 MB, 35k polys, KTX2)]
```

Performance Budgets

Metric	Target Budget per Model	Target Budget per Scene (500 Items)	Measured Prototype (Primitives)
File Size (Compressed)	< 2.5 MB	< 25 MB total download	**In-memory procedural**
Triangle Count	< 40,000 (LOD0)	< 600,000 total	**6,000 triangles**
Draw Calls	1–3 calls	< 300 calls	**4 calls**
Texture VRAM	< 8 MB	< 80 MB	**< 1 MB**

12. AI Assistant Architecture

The AI layout copilot follows a strict human-in-the-loop review workflow:

1. User enters natural language request (e.g. *"Fix clearance violations"*).
2. Assistant streams explanatory response via Server-Sent Events (SSE).
3. Assistant outputs a structured proposal JSON conforming to `AIProposal` schema.
4. Editor renders `<GhostPreview />` displaying proposed positions in translucent green.
5. User chooses **Accept** or **Reject**.
6. On acceptance, the layout modification is applied and recorded as an undoable state snapshot.

13. Performance and Quality Strategy

- **CI/CD Performance Budgets:** Automated Lighthouse and bundle-analyzer checks in GitHub Actions fail builds exceeding 500 KB gzipped initial JS.
- **Telemetry Probes:** Real-time FPS, draw calls, and memory monitoring via Three.js WebGL statistics.
- **Hardware Validation:** Benchmarked on target corporate profile (Apple M1, 8 GB RAM, 1080p viewport).

14. End-to-End Interaction Sequence

Complete Sales Engineer Workflow Scenario

1. Sales engineer drags a Cabinet from palette to floor (5.0, 5.0).
2. Editor snaps position to nearest 0.5m grid increment.
3. Engineer places a Drive at (5.4, 5.0) (< 0.6m clearance).
4. `RuleEngine` recalculates AABB separation and flags a clearance violation (0.2m < 0.6m).

5. Viewport renders red warning borders around both units; Violations Panel displays error.
6. Engineer prompts AI Assistant: *"Fix clearance violation"*.
7. Assistant streams response and proposes moving Drive to (6.5, 5.0).
8. Viewport displays green `<GhostPreview />` at (6.5, 5.0).
9. Engineer clicks **Accept**. Drive moves, clearance violation clears, and BOM updates.
10. Engineer presses `Ctrl+Z` to undo the change; drive returns to (5.4, 5.0) and violation re-appears.

15. Trade-offs and Rejected Alternatives

1. Snapshot Undo vs. Operational Transform (OT):

• *Choice:* Snapshots.

• *Rationale:* Memory footprint for 500 items is under 25 KB per snapshot. OT adds high complexity without tangible benefit for local editing.

2. Tailwind CSS vs. Heavy UI Component Library:

• *Choice:* Tailwind CSS with custom industrial semantic tokens.

• *Rationale:* Eliminates runtime CSS-in-JS overhead while ensuring responsive, accessible layouts.

3. Client-Side vs. Server-Side Clearance Validation:

• *Choice:* Client-side AABB validation for instant feedback + asynchronous server verification for multi-user compliance.

16. Assumptions and Technical Risks

• **Assumptions:** Host application supplies valid OIDC user context via postMessage; CAD files can be pre-processed into glTF 2.0; users have WebGL 2.0 capable browsers.

• **Risks & Mitigations:**

• *VPN Disconnection:* Mitigated by local draft caching in IndexedDB and automatic query reconnection backoff.

• *Integrated GPU Thermal Throttling:* Mitigated by LOD switching, geometry instancing, and dynamic resolution scaling.

17. Testing Strategy and Acceptance Matrix

• **Unit Testing (Vitest):** 100% coverage on clearance distance calculations, AABB rotations, coordinate grid snapping, and undo/redo state transitions.

• **Component Testing:** Properties panel input validation, BOM aggregation tables, language toggle updates.

• **Stress Testing:** Automated 500-item scene population verifying ≥ 50 FPS and < 300 draw calls.